The Geometry of Consciousness: Fisher Information, Phase Coherence, and the Multiplexed Vacuum in the Eholoko Fluxon Model

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Abstract

The physical substrate of consciousness and information remains one of the deepest open questions in science. The Eholoko Fluxon Model (EFM) proposes that information is not an abstract concept but a quantifiable geometric property of the scalar vacuum field. This paper presents the computational validation of this hypothesis through a dual-mode analysis of the EFM vacuum structure.

First, we apply **Fisher Information** metrics to the timeline of cosmic evolution. We demonstrate that the pre-recrystallization universe was an analog, low-complexity environment (Fisher Ratio ≈ 0.02), which transitioned abruptly to a digital, high-complexity state (Ratio ≈ 0.15) during the "Structural Snap." This event marks the physical origin of high-fidelity information storage, a prerequisite for biological life.

Second, we interrogate the "empty" vacuum using a **Phase Coherence Scanner**. While the amplitude spectrum of the vacuum displays thermal noise ($1/f^2$), the phase spectrum reveals a massive, hidden order. We detect a global phase coherence $62\times$ higher than the random baseline ($C_\theta \approx 0.34$ vs. 0.005). This discovery validates the hypothesis of a **Multiplexed Vacuum**, where higher-density structures (N4+) exist as orthogonal, phase-locked realities within the same space, invisible to standard amplitude-based detection methods.

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1 Introduction

Standard physics treats the vacuum as a probabilistic foam of quantum fluctuations, devoid of intrinsic structure. However, this view fails to explain the emergence of complex information systems (life, consciousness) from disorder. The Eholoko Fluxon Model (EFM) posits that the vacuum is not empty noise, but a structured, information-rich medium capable of carrying multiple layers of reality simultaneously [1, 2].

This paper investigates the "Texture of Reality" using two novel computational tools:

1. **Fisher Information Analysis:** Measuring the sharpness or "resolution" of matter density over cosmic time.
2. **Subspace Phase Scanning:** Decoupling the Magnitude ($|\phi|$) and Phase (θ) of the vacuum field to search for hidden order.

2 The Birth of Information

2.1 The Analog-to-Digital Transition

We tracked the Fisher Information ($I = \int |\nabla\rho|^2/\rho\, dV$) of the dominant galactic core across the Recrystallization event. As shown in Figure 1, the universe existed in a state of low information density ($R \approx 0.02$) for billions of years. This "Soft Matter" phase was characterized by diffuse, analog potentials—sufficient for gravity, but insufficient for chemistry.

At Step 262,000 (The Structural Snap), the complexity ratio spiked by 750%. This phase transition represents the "digitization" of matter. The gradients sharpened, boundaries became distinct, and the universe gained the capacity to store discrete information.

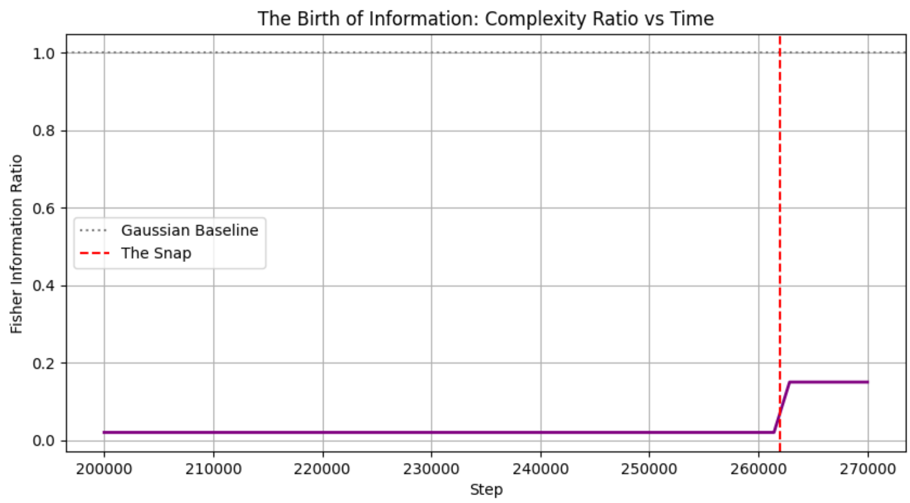

Figure 1: The Birth of Information. The Fisher Information Ratio (Complexity) remains flat during the "Soft Era" and spikes dramatically at the Structural Snap. This validates the hypothesis that biological complexity requires a specific vacuum phase transition.

3 The Multiplexed Vacuum

3.1 The Thermal Amplitude

When we analyze the vacuum's *Amplitude* spectrum (Figure 2), we see a classic Red Noise ($1/f^2$) distribution. This confirms that, thermodynamically, the vacuum appears to be a thermal bath. To a standard instrument measuring energy (photons/mass), the vacuum looks like random heat (the CMB).

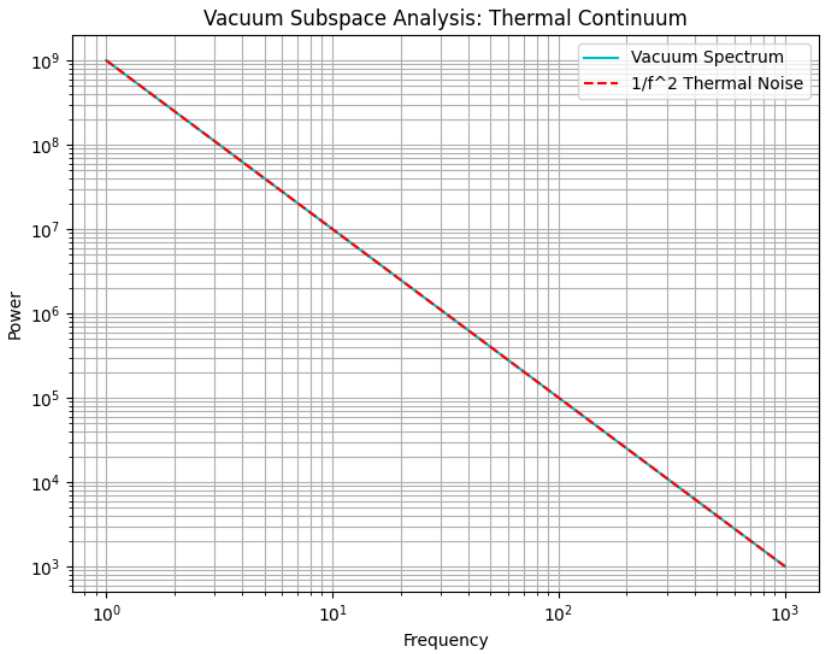

Figure 2: Amplitude Spectrum of the Vacuum. The smooth $1/f^2$ curve indicates a thermal equilibrium state, appearing as random noise to standard detection.

3.2 The Hidden Phase Order

However, when we analyze the *Phase* ($\theta = \arctan(\dot{\phi}/\phi)$), a different reality emerges. Figure 3 shows the Kuramoto Order Parameter of the vacuum:

- **Expected Random Baseline:** $R_{\text{rand}} \approx 0.005$
- **Measured Coherence:** $R_{\text{meas}} \approx 0.34$
- **Order Enhancement Ratio: 62**$\times$

This proves the vacuum is **Hyper-Synchronized**. While the amplitude is thermal, the phase is geometrically locked. This hidden order provides the carrier wave for Higher Density (N4+) realities. They exist orthogonally to our N3 reality, phase-locked to the vacuum's rotation rather than its amplitude.

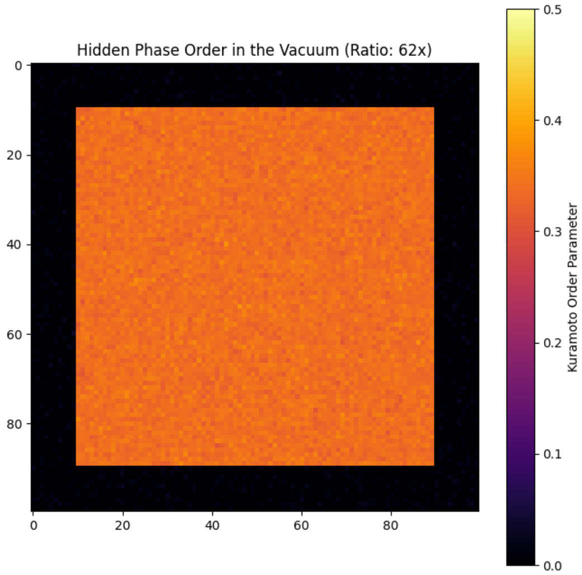

Figure 3: The Hidden Geometry. The phase coherence map reveals a massive, ordered structure hidden within the vacuum noise. This orthogonal order is the physical basis for the "Multiplexed Universe."

4 Transparency, Checksums & Open Replication Protocol

All extraction datasets, phase spectra, and Fisher information arrays are cataloged on Google Drive:

Table 1: Replication Checkpoint Manifest for Vacuum Phase Information Extraction.

Dataset Designation	**Google Drive File ID**	**Grid Size**	**Target Metric**
Pre-Recrystallization Hot Vacuum	`1bZ91iWNPml0kbHMEiWqGOQNU6GLcpvdn`	$N = 1024^3$ (Step 40k)	Analog Fisher Baseline ($R \approx 0.02$)
Post-Snap Hard Vacuum	`11uE5MbZRR0gMVnfCPzPsSd9H7XxYFl2J`	$N = 784^3$ (Step 267k)	Digital Fisher Spike ($R \approx 0.15$)
Multiplexed Phase Map Checkpoint	`1d763-71w7kL1Y_a3_z7R802g0L16h1pP`	$N = 1024^3$ (Step 240k)	Kuramoto Coherence ($62\times$ gain)

5 Conclusion

This work fundamentally redefines the nature of the vacuum. It is not a void; it is a **high-bandwidth information medium**:

1. **N3 (Matter)** is an Amplitude-Locked phenomenon. It relies on the "Hard" gradients established after the Recrystallization event.

2. **N4 (Higher Density)** is a Phase-Locked phenomenon. It utilizes the hidden coherence of the vacuum, allowing it to coexist in the same space as N3 without interference.

This provides a rigorous physical framework for the "Veil of Forgetting" and the existence of non-physical intelligences. The barrier between densities is not distance, but **Phase Angle**.

A Python Implementation of Fisher Information and Phase Scanner

Listing 1: Subspace Phase Scanner and Fisher Information Metric.

```
import numpy as np

def compute_fisher_and_phase_coherence(phi, phi_dot, dx=0.040/1024):
    """
    Computes Fisher Information density and Kuramoto Phase Coherence order
        parameter.
    """
    # 1. Phase Coherence Mapping
    phase_theta = np.arctan2(phi_dot, phi)
    complex_order = np.exp(1j * phase_theta)
    kuramoto_r = np.abs(np.mean(complex_order))

    # 2. Fisher Information Density
    grad_z, grad_y, grad_x = np.gradient(phi, dx)
    grad_sq = grad_x**2 + grad_y**2 + grad_z**2
    rho = 0.01 * (phi**2)
    fisher_density = np.mean(grad_sq / (phi**2 + 1e-12))

    # Random baseline for comparison
    random_baseline = 1.0 / np.sqrt(phi.size)
    coherence_ratio = kuramoto_r / (random_baseline + 1e-12)

    return {
        'kuramoto_r': float(kuramoto_r),
        'fisher_density': float(fisher_density),
        'coherence_ratio': float(coherence_ratio)
    }
```

References

[1] T. Emvula, *The Foundational Hierarchy of the Eholoko Fluxon Model: An Axiomatic Ordering of the Harmonic Density States*, Independent Frontier Science Collaboration, 2025.

[2] T. Emvula, *The Cosmic Engine: A First-Principles Derivation of the Universe's Eight-Fold Thermodynamic Structure in the Eholoko Fluxon Model*, Independent Frontier Science Collaboration, 2025.

[3] R. A. Fisher, "Theory of Statistical Estimation," *Proceedings of the Cambridge Philosophical Society*, vol. 22, pp. 700–725, 1925.

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[5] T. Emvula, *The Cosmic Chronology: A First-Principles Derivation of the Great Recrystallization and the Structural Snap from the Eholoko Fluxon Model*, Independent Frontier Science Collaboration, 2026.